

Asymptotic Analysis of $f(n) = 4n + 40$

Given Function

Consider

$$f(n) = 4n + 40$$

and let

$$g(n) = n.$$

We will verify the function using:

- Big-O notation
- Big- Ω notation
- Big- Θ notation

1. Big-O Analysis

The definition of Big-O is

$$f(n) \leq Cg(n), \quad n \geq n_0$$

Given:

$$f(n) = 4n + 40$$

$$g(n) = n$$

and

$$C = 5.$$

Therefore,

$$4n + 40 \leq 5n$$

Subtracting $4n$ from both sides,

$$40 \leq n.$$

Therefore,

$$n \geq 40$$

Hence,

$$n_0 = 40$$

Therefore,

$$4n + 40 = O(n)$$

with

$$C = 5, \quad n_0 = 40.$$

Verification for Different Values of n

We compare

$$f(n) = 4n + 40$$

with

$$Cg(n) = 5n.$$

The Big-O condition is

$$f(n) \leq 5n.$$

n	$f(n) = 4n + 40$	$g(n) = n$	$5g(n) = 5n$	Condition
1	44	1	5	No
2	48	2	10	No
10	80	10	50	No
20	120	20	100	No
25	140	25	125	No
40	200	40	200	Yes
41	204	41	205	Yes

Therefore, the inequality starts becoming true at

$$n = 40.$$

Thus,

$$f(n) = O(n)$$

with

$$C = 5, \quad n_0 = 40.$$

2. Big- Ω Analysis

The definition of Big- Ω is

$$f(n) \geq cg(n), \quad n \geq n_0.$$

Choose

$$c = 4.$$

Then,

$$4n + 40 \geq 4n.$$

Since

$$40 \geq 0,$$

the inequality is true for every

$$n \geq 1.$$

Therefore,

$$\boxed{n_0 = 1}.$$

Hence,

$$\boxed{4n + 40 = \Omega(n)}$$

with

$$\boxed{c = 4, \quad n_0 = 1}.$$

Verification

$$f(n) = 4n + 40$$

and

$$cg(n) = 4n.$$

n	$f(n) = 4n + 40$	$4g(n) = 4n$	Condition
1	44	4	Yes
2	48	8	Yes
10	80	40	Yes
20	120	80	Yes
25	140	100	Yes
40	200	160	Yes
41	204	164	Yes

Therefore,

$$\boxed{4n + 40 = \Omega(n)}.$$

3. Big- Θ Analysis

The definition of Big- Θ is

$$c_1g(n) \leq f(n) \leq c_2g(n), \quad n \geq n_0.$$

Choose

$$c_1 = 4$$

and

$$c_2 = 5.$$

Therefore,

$$4n \leq 4n + 40 \leq 5n.$$

The left inequality,

$$4n \leq 4n + 40,$$

is always true.

For the right inequality,

$$4n + 40 \leq 5n,$$

we get

$$n \geq 40.$$

Therefore,

$$\boxed{n_0 = 40}.$$

Hence,

$$\boxed{4n + 40 = \Theta(n)}$$

with

$$\boxed{c_1 = 4, \quad c_2 = 5, \quad n_0 = 40}.$$

4. Summary

Notation	Constants	n_0	Result
Big-O	$C = 5$	40	$O(n)$
Big- Ω	$c = 4$	1	$\Omega(n)$
Big- Θ	$c_1 = 4, c_2 = 5$	40	$\Theta(n)$

Therefore,

$$4n + 40 = O(n)$$

$$4n + 40 = \Omega(n)$$

and

$$4n + 40 = \Theta(n)$$

Since the linear term $4n$ dominates the constant 40 as n grows,

$$4n + 40 = \Theta(n).$$

5. Graphical Verification of Big-O

The graph below compares

$$f(n) = 4n + 40$$

with

$$5g(n) = 5n.$$

The two lines intersect at

$$n = 40.$$

For $n \geq 40$,

$$f(n) \leq 5g(n).$$

